

Transscleral local resection versus iodine brachytherapy for uveal melanomas that are large because of tumour height

Ilkka Puusaari · Bertil Damato · Tero Kivelä

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Abstract

Background To compare visual outcome and ocular complications of transscleral local resection (TSR) with those of iodine-125 plaque brachytherapy (IBT) for uveal melanomas categorised as large because of tumour height.

Methods Retrospective study of 87 patients with a uveal melanoma ≤ 16 mm by largest basal diameter (median, 12.6 vs 14.0 mm for TSR and IBT, respectively) and large by height (median, 11.0 vs 10.6 mm) by the Collaborative Ocular Melanoma Study (COMS) criteria. Thirty-three patients underwent TSR in the United Kingdom and 54 underwent IBT in Finland. Loss of 20/65 and 20/400 vision, local tumour recurrence, cataract, iris neovascularization, glaucoma, maculopathy, optic neuropathy, persistent retinal detachment, and vitreous haemorrhage were analysed using competing risks and logistic regression to control for confounders.

Results All patients save one managed with TSR lost 20/70 vision within 3 years. The 2-year cumulative incidence of losing 20/400 vision was 60% (95% CI, 35–75) for TSR and 75% (95% CI, 59–86) for IBT. The 5-year incidence of local tumour recurrence was 41% (95% CI, 17–63) after TSR and 7% (95% CI, 2–17) after IBT. Glaucoma and optic neuropathy were rare after TSR. Cataract, maculopathy, retinal detachment, and vitreous haemorrhage were

common after either treatment. The number of patients needed to treat with TSR instead of IBT was four for one additional patient to benefit by avoiding loss of visual acuity 20/400 for at least 2 years, and the corresponding number was three for one additional patient to experience a local recurrence from TSR.

Conclusions If TSR is further evaluated as an alternative to IBT in avoiding blindness of an eye with a ciliochoroidal melanoma categorised as large by COMS criteria because of its height, special emphasis must be given to increased risk of local tumour recurrence.

Keywords Uveal melanoma · Iodine brachytherapy · Local resection

Introduction

Iodine brachytherapy (IBT) is currently the most widely used eye-conserving treatment option for large uveal melanomas [14, 25, 26, 32]. It is more readily available than charged-particle therapy and gamma knife and is easy to perform. The local tumour recurrence rate is under 10%, and about 80% of eyes can be conserved even when the tumour is large [30]. However, long-term preservation of useful vision is seldom achieved [2, 23, 29, 31]. Loss of vision after brachytherapy is associated with large tumour height, which translates to a high dose of radiation to adjacent tissues [12, 18–20, 31]. The presence of a large, dying intraocular tumour is also thought to contribute to complications [24].

Non-controlled studies suggest that local transscleral resection (TSR) avoids many complications associated with IBT and may thus yield better preservation of vision [2, 10, 23]. Resection is more demanding of the surgeon and more

I. Puusaari (✉) · T. Kivelä
Department of Ophthalmology,
Helsinki University Central Hospital,
PL 220, Haartmaninkatu 4 C,
FI-00029 HUS Helsinki, Finland
e-mail: ilkka.puusaari@hus.fi

B. Damato
Ocular Oncology Service, St. Paul's Eye Unit,
Royal Liverpool University Hospital,
Liverpool, UK

prone to result in local tumour recurrence; furthermore, the patient must be fit for hypotensive anesthesia [2, 23]. Because of these limitations, radiotherapy is sometimes the only feasible eye-conserving option. Nevertheless, evidence-based data would help choose between these two treatments when a choice exists.

In an analysis of 97 patients with a large ciliochoroidal melanoma managed with IBT, many radiation-related complications were associated with tumour height [29], and we predicted that patients with tumours that are large because of their height rather than diameter might do better after TSR than after IBT [11]. To look for evidence in support of this hypothesis and to estimate the size of the potential difference in visual outcome, we compared TSR with IBT in terms of preservation of vision, local tumour recurrence, and main complications in this subgroup of patients with a large melanoma.

Patients and methods

Aims of the study

The primary aim was to test the hypothesis that TSR yields better preservation of visual acuity than IBT in managing uveal melanomas that are classified as large because of their height rather than basal diameter. A secondary aim was to compare local tumour recurrence and main complications between the treatments in this category of patients.

Inclusion criteria

Eligible for this study were patients with a uveal melanoma that met the COMS criteria of large tumour solely because of tumour height ($LBD \leq 16.0$ mm and height >10.0 mm, or >8.0 mm if the tumour was peripapillary) [6], managed with either primary TSR (with or without adjunctive ruthenium brachytherapy) by a single surgeon or IBT between November 1, 1990, and June 1, 2003 [30]. TSR was performed at St. Paul's Eye Unit, Royal Liverpool University Hospital, Liverpool, United Kingdom, and IBT at Department of Ophthalmology, Helsinki University Central Hospital, Helsinki, Finland. Analogous to COMS, patients with extrascleral tumour extension larger than 2 mm, diffuse or multifocal tumours or tumour involvement of the iris were excluded [6, 30].

Enucleation was routinely offered as an alternative to all patients with a large uveal melanoma in both study centres, and recommended to those who had a tumour that was considered too large for eye-conserving therapy, and to those with an extrascleral tumour extension over 2 mm, a diffuse tumour, a ring melanoma, and complications such as choroidal detachment at diagnosis, which are associated

with poor visual prognosis. TSR was not performed on patients who were unfit for hypotensive anaesthesia because of, for example, uncontrolled hypertension, liver or renal disease, cerebrovascular disease or ischaemic heart disease. TSR was usually not recommended if vision was already poor at diagnosis. The elements of informed consent and counselling of the patients have been previously described in detail [10, 30]. The study followed the Declaration of Helsinki.

Clinical evaluation

The variables recorded at baseline included tumour height and largest basal diameter (LBD), evaluated with indirect ophthalmoscopy and B-scan ultrasonography, best corrected visual acuity, and intraocular pressure (IOP). Tumour margins, the distance from the posterior margin to the foveola and optic disc, and the presence of retinal detachment and vitreous hemorrhage were assessed with biomicroscopy using a +90.0 diopter convex lens (Volk Optical, Mentor, Ohio) and binocular indirect ophthalmoscopy. Gonioscopy was used to determine the anterior tumour margin and to detect neovascularization of the iris and chamber angle.

Treatment and follow-up

The techniques used for transscleral local resection have evolved over time [7, 10]. The resections were performed by the same surgeon and under hypotensive anaesthesia to minimise haemorrhage. Briefly, a partial-thickness scleral flap was prepared around the tumour, which was resected together with the deep scleral lamella and a safety margin of surrounding normal choroid, if possible. Adjunctive ruthenium-106 brachytherapy (RuBT; 100 Gy to 2–3 mm) was administered at the end of the procedure to 20 patients, all but one of them treated after 1995. When adjunctive RuBT is used, it is possible to resect a small extrascleral extension before creating the scleral flap.

IBT was administered using custom-made, non-collimating gold plaques and a previously described protocol [23, 28, 30]. The prescription dose was initially 100 Gy to the tumour apex as a rule, and 80 Gy if the tumour was very thick in an effort to limit complications. Since 1997, the prescription dose has been 80 Gy to the apex, and very thick tumours have received a prescription dose of 60–70 Gy [30]. Median dose to tumour apex was 81 Gy (range, 42–105). Median dose rate at tumour apex was 53 cGy/h (range, 14–169) and median treatment duration was 153 h (range, 58–459).

In both study centres, patients were prospectively assessed at approximately 1 month after treatment, at 3–6 months, at every 6 months for 3–6 years, and at least once a year thereafter, in part at referring hospitals. Because

of differences in health care systems, total follow-up times were notably shorter in the TSR arm (median, 2.3 vs 5.3 years; $P=0.0004$ Kruskal-Wallis test). Some of the TSR patients were from overseas and had, consequently, short follow-up at the study centre.

Outcome variables

Time to the following outcomes had been prospectively recorded in computerised databases of both study centres:

1. Loss of visual acuity levels of 20/70 and 20/400 (corresponding to loss of reading vision and blindness of the study eye by the World Health Organisation criteria, respectively).
2. Local recurrence, defined as an unequivocal 1.0 mm increase in lateral tumour dimension or tumour thickness, classified as marginal and vertical recurrence, respectively [22], or appearance of a new tumour in a previously uninvolved area.

3. Secondary enucleation.
4. Progression of lens opacity, typically initially posterior or subcapsular cataract in type, which decreased or would have decreased visual acuity if vision was reduced because of other reasons, evaluated by slit lamp biomicroscopy.
5. Neovascularization of the iris, chamber angle or both, evaluated by slit lamp biomicroscopy and, if suspicious vessels were seen or IOP was elevated, by gonioscopy.
6. Glaucoma, defined as IOP over 24 mmHg as measured by applanation tonometry and either confirmed at the next follow-up examination or considered an indication for treatment.
7. Maculopathy, categorised as scarring, macular pucker, cystic macular oedema, or radiation retinopathy (retinal hemorrhages, microaneurysms, microinfarcts, oedema or exudation), evaluated by slit lamp biomicroscopy using a noncontact or contact lens.
8. Optic neuropathy, categorised as radiation papillopathy (oedema, peripapillary splinter haemorrhages or microinfarcts) or optic atrophy.

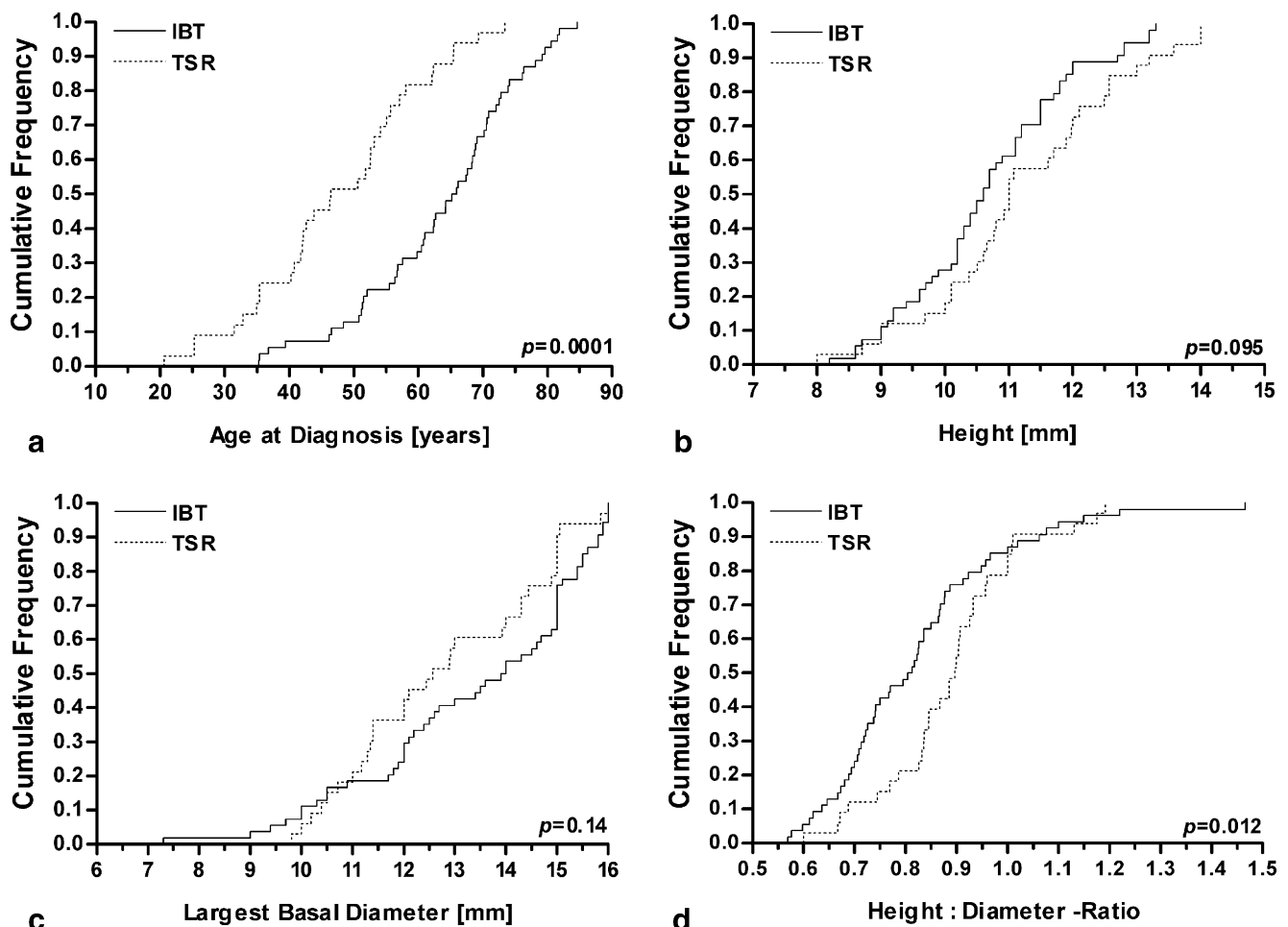


Fig. 1 Cumulative frequency distribution of **a** age at diagnosis, **b** tumour height, **c** largest basal diameter, and **d** tumour H:D ratio of 33 patients managed with TSR, 20 of whom also received adjuvant

RuBT, and 54 patients managed with IBT, who had a uveal melanoma large by the COMS criteria only because of its height

9. Retinal detachment, involving at least one quadrant, which first occurred or increased after treatment and either persisted for 6 months or was considered an indication for surgery, evaluated by indirect ophthalmoscopy or by ultrasonography after visibility to the fundus was compromised. Transient retinal detachment that resolved spontaneously and localised retinal detachment directly over or around the tumour were disregarded.
10. Vitreous haemorrhage occurring or recurring after treatment, evaluated by binocular indirect ophthalmoscopy.

Statistical methods

Follow-up data were analysed with Stata statistical software (release 7.0, Stata, College Station, Tex.) with the -stcompet-automatic do-file (Boston College Statistical Software Components Archive, available at <http://ideas.repec.org/c/boc/bocode/s431301.html>) and R statistical software (version 1.4.0, The R Foundation for Statistical Computing, Vienna, Austria; available at <http://www.r-project.org>) with the -cmprsk- library.

Cumulative frequencies of key baseline characteristics, patient age, tumour height, tumour LBD and height:LBD ratio were plotted for both study arms and compared using Kruskal-Wallis test. Cumulative frequency of visual acuity levels at baseline and at 1, 2, and 3 years after diagnosis was also plotted and compared similarly.

Analysis of visual outcome Cumulative incidence instead of Kaplan-Meier analysis was chosen to assess time to loss of visual acuity in order to account for death as a competing risk [16, 27]. Because competing risks regression, as currently implemented in available software, cannot be used to appropriately analyse counting process type data, multiple-failure models for visual acuity loss were not constructed. Moreover, an exploratory Cox proportional hazards regression model for visual acuity loss did not fulfil the assumption of proportional hazards [33]. Instead, logistic regression models which included treatment type and patient age, tumour height-to-diameter (H:D) ratio and baseline visual acuity as confounders, were used to analyse visual acuity loss at 1, 2, and 3 years from treatment.

Analogous to COMS, mean visual acuity, after transformation to a logarithmic scale, was plotted for each treatment arm at 6-month intervals [25]. Measurements below 20/1,600, such as counting fingers or hand movements, were set equal to 20/1,600. Enucleated eyes and eyes with no light perception were counted as having 20/2,500 vision. If the patient did not have a measurement

available at all appropriate intervals, the visual acuity was taken to be the average of the previous and the next visit. In case of a longer gap in follow-up, the patient was censored from the analysis.

The number of patients needed to treat with TSR instead of IBT for one additional patient to benefit (NNTB) by maintaining 20/400 vision or better for at least 2 years from treatment was calculated with its 95% confidence interval (CI) using the Wilson score method [1, 3]. This method was also used to estimate the harm from local tumour recurrence were the patients managed with TSR instead of IBT. A negative number needed to treat indicates that one additional patient will be harmed (NNTH) by the treatment.

Analysis of recurrence and complications Time to local tumour recurrence and individual complications was assessed by cumulative incidence analysis with death as a competing risk. Enucleation was not handled as a competing risk when analysing complications because of different

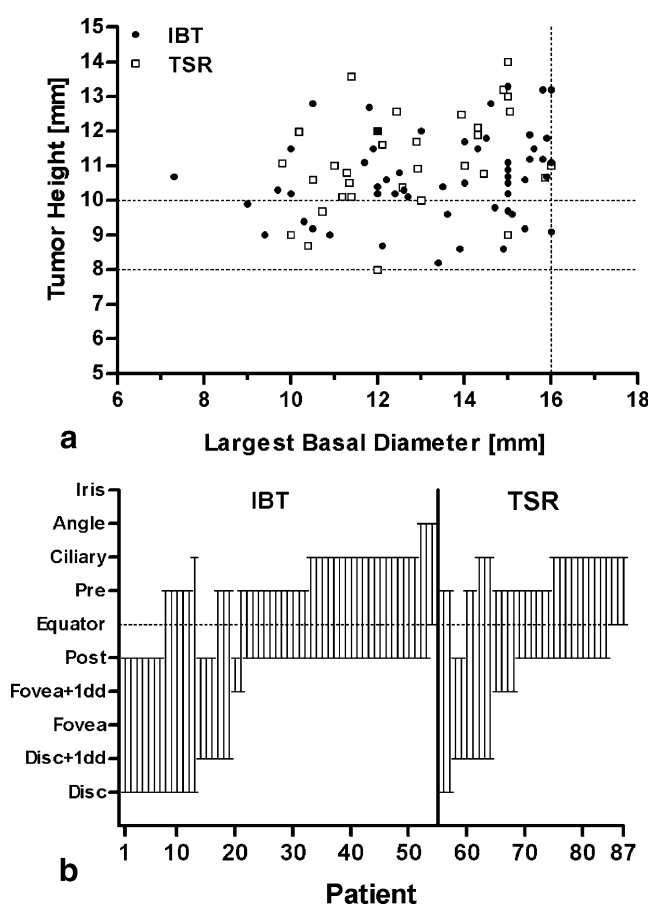


Fig. 2a, b Characteristics of 87 uveal melanomas that were large by the COMS criteria only because of their height, of which 33 were managed with TSR and 54 with IBT. **a** Scatterplot of largest basal diameter against tumour height. *Dashed lines* indicate the COMS criteria for a large tumour. **b** Plot of tumour location. *Bars* indicate the position of the anterior and posterior margin of each tumour

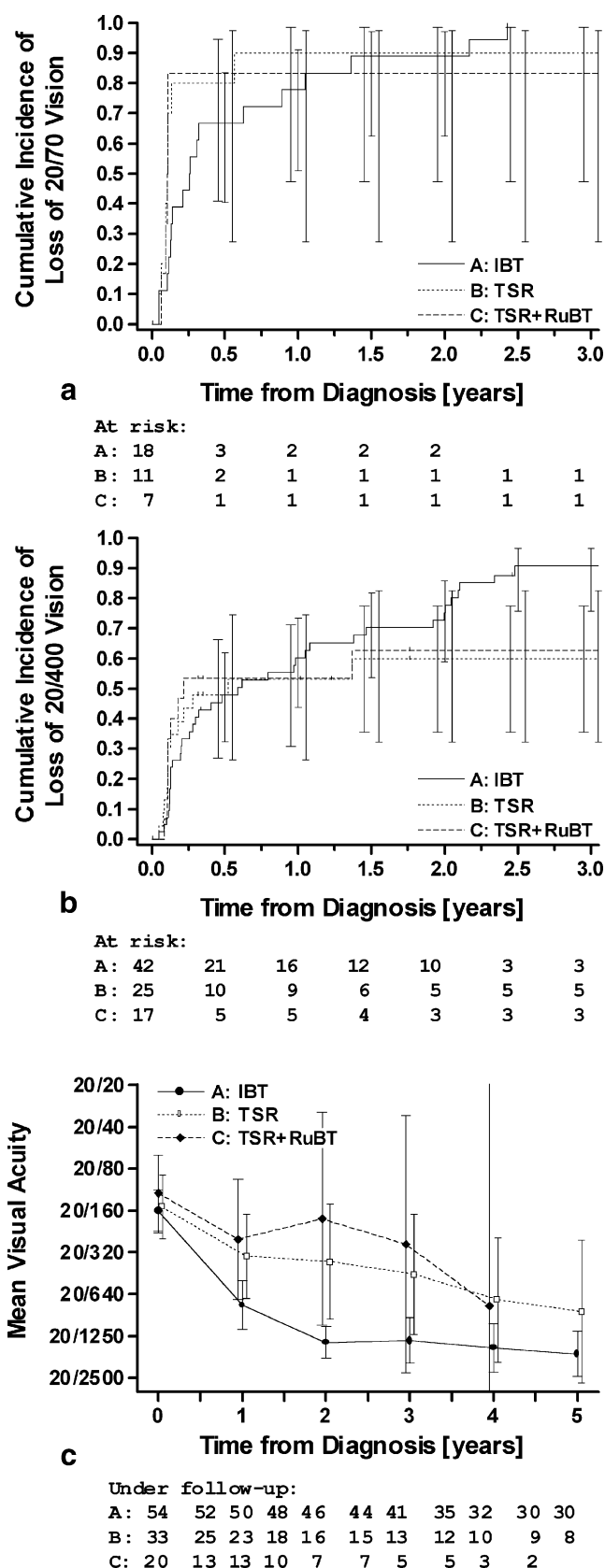


Fig. 3 Cumulative incidence of first loss **a** 20/70 vision and **b** 20/400 vision of 33 patients managed with TSR, 20 of whom also received adjuvant RuBT, and 54 patients managed with IBT, who had a large uveal melanoma, which was large according to the COMS criteria only because of its height. **c** Plot of mean visual acuity during follow-up. Bars show 95% CIs

indications for enucleation in the participating centres. Otherwise, enucleation would have introduced bias by decreasing the cumulative incidence of adverse events observed in the study centre which more readily performed secondary enucleation. Patients were censored from the analysis at the time of secondary enucleation or at the end of follow-up and, as regards posterior segment complications (maculopathy, optic neuropathy, vitreous haemorrhage, and retinal detachment), when visibility to the fundus was lost. After censoring, the patients are still regarded as being at risk of the event of interest [16].

Competing risks proportional hazards regression was used to adjust for tumour and host factors associated with time to each complication [15]. In addition to treatment type and baseline variables that were significantly different between the treatment arms, additional variables were included in the model from those explanatory variables that are known to be associated with the complication of interest [21]. The number of variables in the final model was limited according to the number of events, based on a rule that there had to be at least 15–20 events for each additional variable [27]. The prevalence of potentially reversible complications, vitreous haemorrhage and retinal detachment, was assessed by calculating the proportion of patients with the complication at 6-month intervals. Analogous to COMS, patients with missing data were considered to have a complication if it was present at the previous and a subsequent visit no more than 6 months later [25]. A patient with longer data gaps was censored from the calculation at that time point.

Results

The patients in the TSR arm were, on average, 20 years younger than those in the IBT arm (mean age at diagnosis, 46 vs 66 years, $P=0.0001$, Kruskal-Wallis test; Fig. 1a). One patient in the TSR arm and 18 patients in the IBT arm were over 70 years old. Three patients in the IBT arm had diabetes without retinopathy. The tumours tended to be thicker in the TSR arm (median height, 11.0 mm; range, 8.0–14) compared with the IBT arm (median, 10.6 mm; range, 8.2–13.3; $P=0.095$; Fig. 1b). The median LBD was 12.5 mm (range 9.8–16) in the TSR arm and 14.0 mm (range, 7.3–16) in the IBT arm ($P=0.14$; Fig. 1c) and the tumour H:D ratio was higher in the TSR arm (median 0.90 vs 0.81, $P=0.012$; Fig. 1d).

The anterior tumour margin involved the ciliary body in 24% of TSR patients and in 39% of those managed with IBT patients ($P=0.24$, Fisher's exact test, two-sided; Fig. 2). The posterior tumour margin extended to within 2 mm of the optic disc in no eye managed with TSR compared with 39% of the eyes managed with IBT ($P<0.001$), and 24% vs 20% of tumours were located within 2 mm of the fovea ($P=0.79$), respectively. One patient in the TSR arm had a small extrascleral extension present at diagnosis but no recurrence after 6.8 years of follow-up.

Loss of visual acuity

At baseline, 11 patients (33%) in the TSR arm and 18 patients (33%) in the IBT arm had visual acuity better than 20/70. The 2-year cumulative incidence of losing 20/70 vision was 90% (95% CI, 47–99) in the TSR arm and 89% (95% CI, 62–97) in the IBT arm. All patients managed with IBT lost this level of vision within 30 months, but one regained it after cataract surgery. One patient in the TSR

arm retained this level of visual acuity (Fig. 3a) for at least 4 years. Baseline visual acuity was better than 20/400 in 25 patients (75%) of the TSR arm and in 42 patients (78%) of the IBT arm. After TSR, the cumulative incidence of loss of visual acuity 20/400 was 53% (95% CI, 31–71) at one and 60% (95% CI, 35–77) at 2 and 3 years (Fig. 3b). After IBT, the corresponding incidences were 60% (95% CI, 44–73), 75% (95% CI, 59–86) and 91% (95% CI, 76–97).

The mean visual acuity after TSR was 20/320 to 20/640 throughout the first five years and approximately one line better than the mean visual acuity after IBT, which varied between 20/640 and 20/1,250 (Fig. 3c). The cumulative frequency plot of visual acuity was comparable at baseline (Fig. 4a) but at one, two, and three years differed significantly in favour of TSR (Fig. 4b–d).

A multivariate logistic regression model was used to adjust for baseline visual acuity, age, and H:D ratio. Treatment with TSR was not consistently associated with avoiding loss of 20/70 vision. Treatment with TSR was independently predictive of avoiding loss of visual acuity

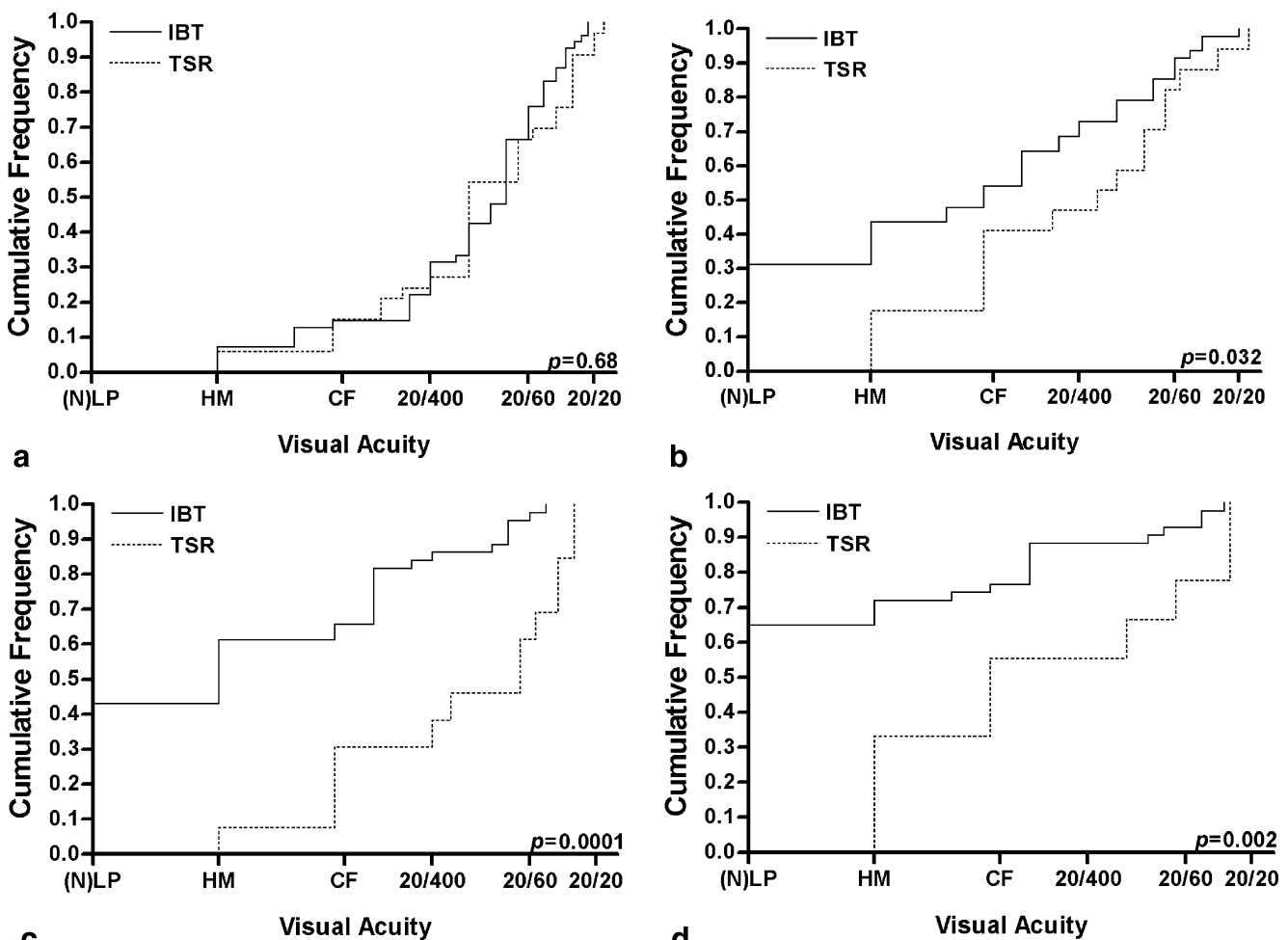


Fig. 4 Cumulative frequency distribution of visual acuity at **a** baseline, **b** 1 year, **c** 2 years, and **d** 3 years after treatment of 33 patients managed with TSR and 54 patients managed with IBT, who had a large uveal

melanoma, which was large according to the COMS criteria only because of its height. Kruskal-Wallis test

20/400 at 1, 2, and 3 years, with the largest difference present at 2 years ($P=0.005$, Table 1).

Local tumour recurrence

Local tumour recurrence developed in nine patients managed with TSR and in three managed with IBT. The 5-year cumulative incidences of local recurrence were 41% (95% CI, 17–63) and 7% (95% CI, 2–17), respectively (Fig. 5a). The incidence was 37% (95% CI, 7–66) in the TSR subgroup that received adjuvant irradiation treatment and 46% (95% CI, 13–74) in the subgroup that did not.

Of the nine patients with recurrence after TSR, six were managed with secondary irradiation (two with RuBT and four with PBT), one with transpupillary thermotherapy, and two underwent enucleation. Three patients managed with secondary radiotherapy experienced a second recurrence,

Fig. 5 Cumulative incidence of **a** local tumour recurrence, **b** development or progression of cataract, **c** glaucoma (plot for both TSR groups are on top of one another at 0), **d** maculopathy, **e** optic neuropathy, and **f** retinal detachment of 33 patients managed with TSR, 20 of whom also received adjuvant RuBT, and 54 patients managed with iodine brachytherapy (IBT), who had a large uveal melanoma large, which was according to the COMS criteria only because of its height. Death was analysed as a competing risk but not plotted. **g, h** The prevalence of retinal detachment and vitreous haemorrhage, respectively. Bars indicate 95% CIs and ticks show censored observations

and then underwent enucleation. Two of the three patients with local tumour recurrence after IBT were managed with secondary IBT and one underwent enucleation.

Enucleation

Eight patients in the TSR arm and five patients in the IBT arm underwent secondary enucleation. The cumulative

Table 1 Logistic regression of loss of visual acuity after IBT and TSR of 87 patients with a uveal melanoma classified as large because of its height by the Collaborative Ocular Melanoma Study criteria (*SE* standard error, *H:D* tumour height to largest basal diameter ratio)

Variable	Loss of reading vision (20/70)				Loss of ambulatory vision (20/400)			
	Coefficient (SE)	Wald chi-square	<i>P</i>	Odds ratio (95% CI)	Coefficient (SE)	Wald chi-square	<i>P</i>	Odds ratio (95% CI)
At 1 year								
Univariate	–2 log likelihood=55.7				–2 log likelihood=83.1			
Treatment ^a	–0.227 (0.756)	0.090	0.76	0.80 (0.18–3.51)	–0.906 (0.577)	2.46	0.116	0.40 (0.13–1.25)
Multivariate	–2 log likelihood=42.6				–2 log likelihood=51.7			
Age ^b	–0.010 (0.032)	0.09	0.76	0.99 (0.93–1.06)	–0.005 (0.027)	0.029	0.87	1.00 (0.94–1.05)
H:D ^c	–0.370 (0.247)	2.25	0.13	0.69 (0.43–1.12)	–0.134 (0.222)	0.360	0.55	0.87 (0.57–1.35)
Baseline VA ^d	–0.155 (0.057)	7.24	0.007	0.86 (0.77–0.96)	–0.236 (0.057)	17.2	<0.001	0.79 (0.71–0.88)
Treatment ^a	–0.302 (1.001)	0.090	0.76	0.74 (0.10–5.26)	–1.752 (0.932)	3.534	0.060	0.17 (0.03–1.08)
At 2 years								
Univariate	–2 log likelihood=33.6				–2 log likelihood=54.6			
Treatment ^a	–2.575 (0.921)	7.78	0.005	0.08 (0.01–0.46)	–2.476 (0.729)	11.56	0.001	0.084 (0.020–0.35)
Multivariate	–2 log likelihood=27.1				–2 log likelihood=34.1			
Age ^b	–0.021 (0.044)	0.23	0.63	0.98 (0.90–1.07)	0.002 (0.035)	0.003	0.96	1.00 (0.94–1.07)
H:D ^c	–0.115 (0.320)	0.13	0.72	0.89 (0.47–1.68)	–0.182 (0.230)	0.62	0.43	0.83 (0.53–1.31)
Baseline VA ^d	–0.145 (0.068)	4.54	0.03	0.86 (0.76–0.99)	–0.227 (0.072)	10.1	0.001	0.80 (0.69–0.92)
Treatment ^a	–2.872 (1.202)	5.71	0.017	0.06 (0.01–0.60)	–3.532 (1.246)	8.07	0.005	0.03 (0.003–0.34)
At 3 years								
Univariate	–2 log likelihood=31.3				–2 log likelihood=43.3s			
Treatment ^a	–1.338 (1.001)	1.80	0.18	0.26 (0.04–1.87)	–1.805 (0.822)	4.796	0.028	0.16 (0.033–0.82)
Multivariate	–2 log likelihood=22.6				–2 log likelihood=15.4			
Age ^b	0.006 (0.049)	0.02	0.90	1.01 (0.91–1.11)	–0.064 (0.068)	0.90	0.34	0.94 (0.82–1.07)
H:D ^c	–0.107 (0.368)	0.08	0.77	1.11 (0.54–2.29)	–0.159 (0.349)	0.21	0.65	0.85 (0.43–1.69)
Baseline VA ^d	–0.186 (0.076)	5.90	0.02	0.83 (0.72–0.96)	–0.483 (0.201)	5.76	0.016	0.62 (0.42–0.91)
Treatment ^a	–0.799 (1.322)	0.36	0.55	0.45 (0.03–6.01)	–4.690 (2.605)	3.24	0.072	0.01 (0.0001–1.52)

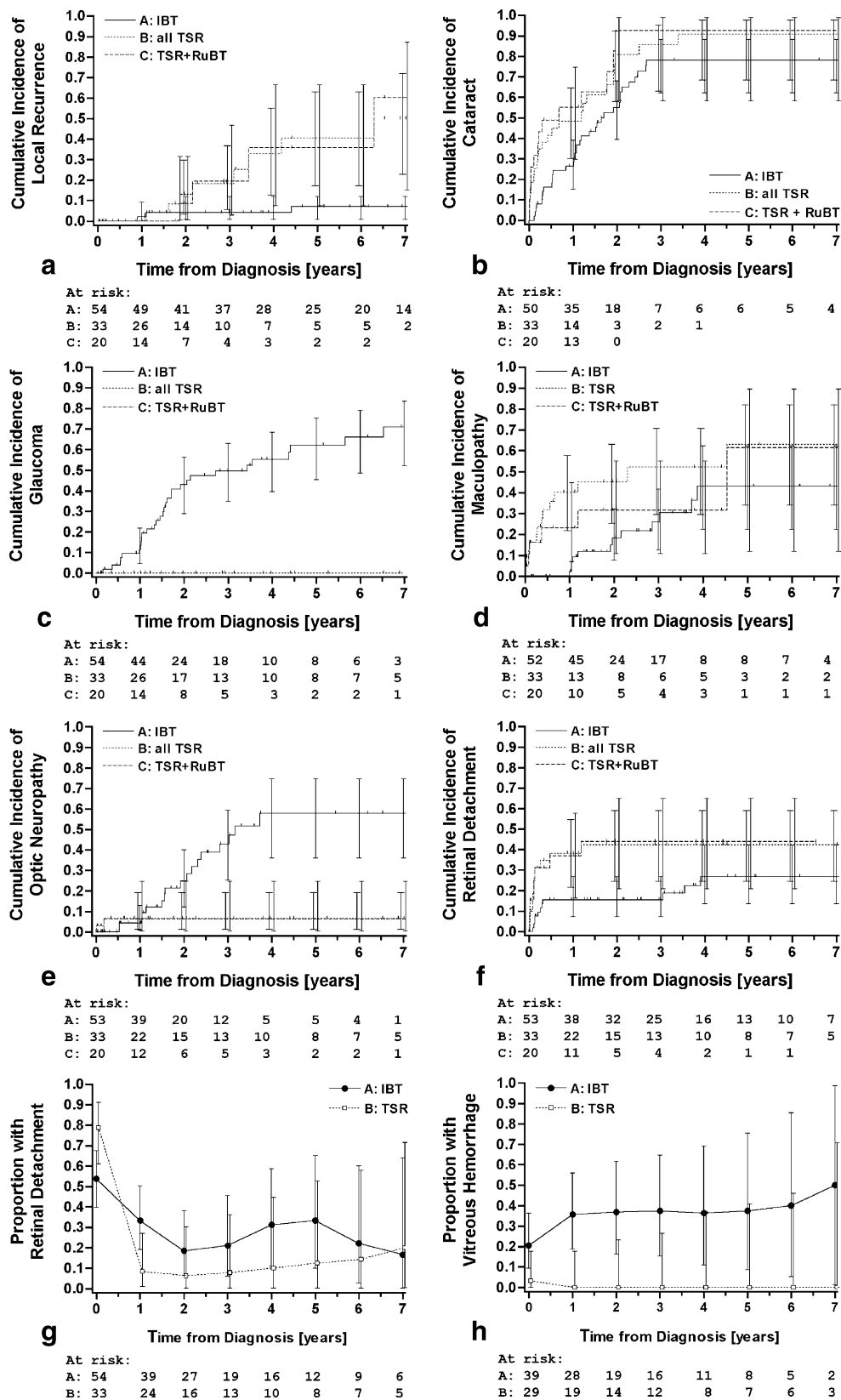
VA=visual acuity

^a Coding: 0=IBT, 1=TSR

^b Continuous variable, years

^c Continuous variable, per 0.1 units

^d Continuous variable, per 0.1 logMAR units



5-year incidence of enucleation was 28% (95% CI, 7–40) and 10% (95% CI 4–20), respectively. Five TSR patients underwent enucleation because of local tumour recurrence, two eyes were removed within 2 months of TSR because of total retinal detachment, and one enucleation was performed in a regional hospital for an unspecified reason.

In addition to the patient who underwent enucleation because of local tumour recurrence, four enucleations were performed after IBT because of intractable neovascular glaucoma.

Cataract

Cataract was present at baseline in 20 (37%) of the eyes managed with IBT and four (7%) of them had previously undergone cataract extraction. All eyes in the TSR arm were phakic and three (9%) had cataract at baseline. By cumulative incidence analysis, cataract developed or progressed within 5 years in 91% (95% CI, 68–98) of the 38 eyes managed with TSR and in 78% (95% CI, 62–88) of 50 phakic eyes managed with IBT (Fig. 5b). Eleven patients in the TSR arm and 17 patients in the IBT arm underwent cataract extraction, and two and three of them, respectively, regained more than two Snellen lines. All but one later lost 20/65 vision.

In a multivariate competing risks regression model, which adjusted for age, baseline lens opacities and tumour height, TSR (HR 1.87, $P=0.060$) was independently associated with shorter time to cataract progression.

Iris neovascularization and glaucoma

Baseline intraocular pressure was over 24 mmHg in one eye in the TSR arm and in two eyes in the IBT arm. Iris neovascularization and glaucoma developed only in patients who underwent IBT (Fig. 5c). The 5-year cumulative incidences were 64% (95% CI, 47–77) and 62% (95% CI, 45–75; Fig. 5c), respectively.

Maculopathy

Maculopathy occurred within 5 years in 63% (95% CI, 34–82) of eyes after TSR and in 43% (95% CI, 22–62) of eyes managed with IBT (Fig. 5d). Retinal folds and preretinal fibrosis were found in seven eyes, cystoid macular oedema in three eyes, and involvement of the macula by the surgical coloboma in four eyes after TSR. Scarring and atrophy of the macula was seen in four eyes and radiation retinopathy in eight eyes after IBT. Adjusting for age and distance from posterior tumour margin to the fovea, treatment with TSR (HR 4.86, $P=0.013$) was independently associated with shorter time to maculopathy.

Optic neuropathy

The 5-year cumulative incidence of optic neuropathy was 7% (95% CI, 1–19) and 58% (95% CI, 36–75) after TSR and IBT, respectively (Fig. 5e). One of the three TSR patients who developed optic neuropathy received RuBT as adjuvant therapy, but the neuropathy resulted from ocular hypotony; one developed a swollen disc soon after surgery, and one presented with optic atrophy 7 years after TSR. Adjusting for distance from posterior tumour margin to the disc, IBT (HR 3.74, $P=0.042$) was an independent predictor of optic neuropathy.

Persistent retinal detachment

Retinal detachment of more than one quadrant was present at baseline in 25 (76%) and 29 (54%) of the eyes in the TSR and IBT arms, respectively (Fig. 5f). New retinal detachment after TSR was rhegmatogenous in type and its 5-year cumulative incidence was 43% (95% CI, 25–59). Of 14 patients with retinal detachment after TSR, nine underwent pars plana vitrectomy, one was managed with scleral buckling, and three patients with total retinal detachment and no visibility to the fundus underwent enucleation. The 5-year cumulative incidence of persistent or recurrent exudative retinal detachment after IBT was 27% (95% CI, 14–42). The prevalence of retinal detachment decreased rapidly to under 20% after TSR but remained 10–20 percentage points higher after IBT (Fig. 5g).

Vitreous haemorrhage

Vitreous haemorrhage was present at diagnosis in one eye (3%) in the TSR arm and in eight eyes (15%) in the IBT arm. By cumulative incidence analysis, 69% (95% CI, 50–82) of patients managed with TSR experienced vitreous bleeding within 6 months of surgery but none had new bleeding after this period. Ten (48%) haemorrhages resolved spontaneously, seven eyes underwent vitrectomy (six had other indications to surgery as well), and three eyes with a persistent haemorrhage and a total retinal detachment underwent enucleation. The 5-year cumulative incidence of vitreous haemorrhage was 39% (95% CI, 23–55) after IBT. One patient underwent vitrectomy, six haemorrhages spontaneously resolved and ten persisted throughout follow-up. The prevalence of persistent or recurrent bleeding after IBT varied from 35% to 40% (Fig. 5h).

Number needed to treat

Of patients with better than 20/400 vision at diagnosis, 13 in the TSR arm and 38 in the IBT arm were under follow-

up for at least 2 years after treatment. Five local tumour recurrences were observed in the TSR arm and two in the IBT arm. In both groups five patients maintained 20/400 vision for at least 2 years (range, 2.5–11). The number needed to treat with TSR instead of IBT for one additional patient to gain this benefit by was 4.0 (95% CI, NNTB 1.9–NNTB 570; Fig. 6). The corresponding number to treat with TSR for one more patient to be harmed by local recurrence was 3.0 (95% CI, NNTH 1.7–NNTH 10.9).

Discussion

Based on the vision endpoints of this study, TSR could be further evaluated as an alternative to IBT in avoiding blindness of an eye with a ciliochoroidal melanoma that is large by COMS criteria only because of its height. Cumulative incidence of blindness, which evaluates the timing of vision loss but lacks the ability to account for recovery of lost vision, suggests that few new cases of blindness occur 1 year after TSR. Mean visual acuity after TSR was 1–2 logarithmic lines better throughout the first

5 years, and logistic regression models at 1, 2 and 3 years, which adjusted for age, baseline visual acuity and tumour H:D ratio identified treatment with TSR as a significant, independent predictor of avoiding blindness of the tumour eye. TSR was also associated with preservation of reading vision at 2 years but this finding was not statistically significant at other assessed time points, possibly because of a smaller difference and fewer patients at risk.

The incidences of neovascular glaucoma and optic neuropathy, two major complications of IBT, were low after TSR and not different from previous series of mainly smaller, unselected tumours [2, 23]. Nevertheless, three patients developed optic neuropathy after TSR, possibly as a result of ocular hypotony and ischemia. Avoiding these complications is likely to be the main explanation for better retention of vision after TSR in our study. Optic neuropathy is associated with a radiation dose over 50 Gy to the optic nerve [17, 28], which was exceeded in all but 11 eyes in the IBT arm.

Radiation injury to ocular vasculature, presence of a large, necrotic intraocular tumour mass, and chronic retinal detachment may contribute to iris neovascularization and neovascular glaucoma in eyes with uveal melanoma [4, 13, 24]. By limiting irradiation to a small adjuvant dose, removing the tumour mass and, usually, the exudative retinal detachment, TSR appears to minimise these complications. The fact that total retinal detachment after TSR that was not amenable to surgery was usually managed with enucleation probably contributes to the small incidence of neovascular glaucoma after TSR in this study.

Cataract, maculopathy, persistent retinal detachment, and vitreous haemorrhage were common even after TSR, which contrasts with previous series of medium- to large-sized tumours [2, 23]. This is likely explained by the uniformly large tumour dimensions in the present series. Large LBD increases the likelihood of macular damage during TSR. Retinal detachment after TSR is typically rhegmatogenous in type, correlates with tumour height, occurs soon after surgery, and can often be managed with a prompt pars plana vitrectomy [8]. Early haemorrhage, which was not uncommon after TSR, usually resolves spontaneously or is cleared with vitrectomy. Despite the high frequency of these complications after TSR, preservation of visual acuity was still better than after IBT [23].

Several limitations must be recognised when interpreting the results of the present study. Uveal melanomas that are classified as large because of their height are uncommon, making it difficult to organise a prospective study with adequate sample size. The retrospective setting used in this study may introduce bias in treatment selection, follow-up methods and baseline variables. We used a two-centre design to limit selection bias in choosing between the two treatments. This, however, introduces differences in follow-

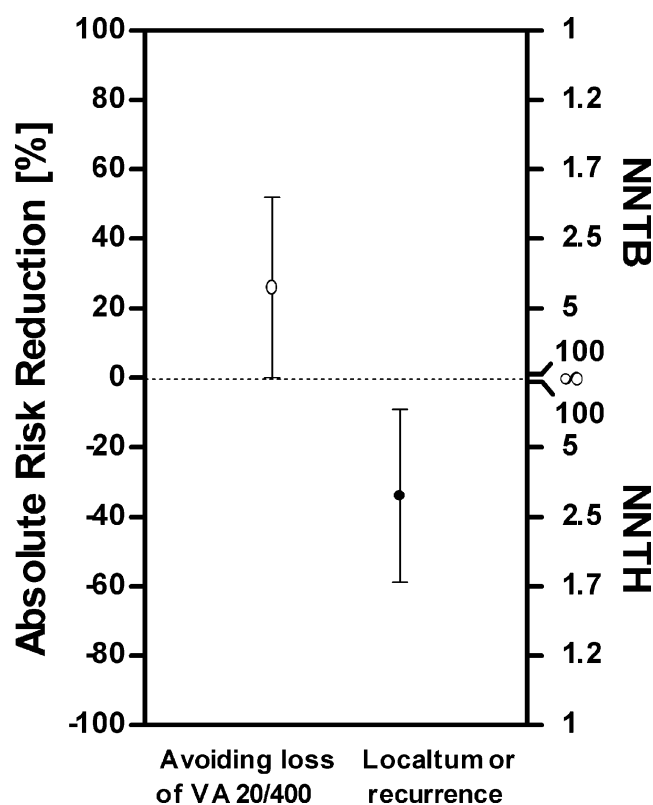


Fig. 6 The number of patients needed to treat with TSR instead of IBT for one additional patient with a uveal melanoma large by the COMS criteria only because of its height to benefit by maintaining 20/400 vision or better for at least 2 years (NNTB), and one additional patient to be harmed by local tumour recurrence (NNTH). Bars indicate 95% CIs

up and indications for secondary enucleation. Eyes managed with TSR in which fundus visibility or vision was lost were enucleated in most cases. After IBT, the policy was to offer enucleation as a safety measure if visibility was lost, but most patients refused this option and preferred to retain the eye for cosmetic reasons [30]. This not only makes the comparison of secondary enucleation rates after the two treatments biased but also risks overestimating any difference in complication rate in favour of TSR. The latter problem was taken into account by censoring enucleated eyes from analysis at the time of enucleation instead of analysing enucleation as a competing risk. This design produces a slight overall overestimation of the incidence of complications in both arms by disregarding the fact that patients are no longer at risk of ocular complications after secondary enucleation [29]. The inability to detect posterior segment complications after fundus visibility was lost was taken into account by censoring patients from analyses at that point.

The differences in tumour size and location as well as patient age at baseline were adjusted for in the regression models. In most models, including the baseline confounders strengthened the association between treatment type and the outcome of interest. The reason why fewer posteriorly located tumours were included in the TSR arm can at least in part be due to the fact that TSR was rarely recommended if there was little vision left to save. Finally, the centre-related difference in follow-up times is considerable. However, the majority of complications occur within the first 3 years after either treatment [2, 23, 29], and cumulative incidence analysis and competing risks regression take the varying lengths of follow-up appropriately into account. The incidence of metastasis and tumour-related mortality were, however, beyond the scope of this study because longer follow-up times would be needed to reliably assess these outcomes.

A greater risk of local recurrence remains a drawback of TSR and must be weighed against the prospect of avoiding blindness in the tumour eye. This risk also emphasises the need for close follow-up by an ocular oncologist after TSR. The apparently small benefit from adjuvant RuBT in this series can be in part attributed to the larger than average size of the tumours. Most recurrences arose from the edges of the resected area which may not have been adequately covered by the 20 mm plaque chosen for this purpose, especially in patients with irregularly shaped tumours and tumours close to the optic disc. Because recurrence after TSR usually results from tumour margins that were not clear, it may not reflect increased malignancy of the tumour. [5] Indeed, more frequent local recurrence has not been linked to increased mortality after TSR compared with IBT [23]. The 5-year recurrence rate was not higher than in eyes with smaller tumours [2, 11, 23,] and the results are

improving with refinements of surgical technique [9, 11]. We estimated that about four additional local recurrences occur after TSR for every three patients prevented from losing 20/400 vision for at least two years.

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